INCH-POUND

MIL-K-71151 (AR)
01 November 1993

MILITARY SPECIFICATION

KIT, REFURBISHMENT, FOR CARTRIDGE, 60MM, PRACTICE, SHORT RANGE, M766 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for the refurbishment kit for Cartridge, 60MM Practice, Short Range, M766 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12953609 - Packing of Box, Fiberboard, Inner for Refurbishment Kit, M766
12953610 - Packing and Marking for Box, Fiberboard, Outer for Refurbishment Kit, M766
12953611 - Packing and Marking for Box, Wood, Wirebound for Refurbishment Kit, M766

QUALITY ASSURANCE PROVISIONS

QAP C12961001 - Dud Plug

(Copies of drawings and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

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3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications. The fuzes (M779), obturators, and ignition cartridges (M987) shall be from lots that meet their respective specification requirements and have been accepted by the Government.

3.2 Obturator. The obturator shall seat flush or below flush when properly assembled to a serviceable carrier round.

3.3 Breech plug firing pin position. The firing pin shall be flush to below flush relative to the rear face of the breech plug after assembly against the propelling charge.

3.4 Fuze striker arming indicator marking. The red arming indicator on the forward section of the fuze striker shall not be visible when the striker is in the safe position and shall be readily visible when the striker is in the armed position.

3.5 Increment plug. The increment plug shall be able to be easily inserted and be firmly seated in the increment plug hole of the projectile body, and be removable with an outward pull of 14 ± 8 pounds on the increment plug pull tab when pulled perpendicular to the projectile body.

3.6 Proving ground test. The refurbished cartridges shall comply with the following requirements when tested in accordance with 4.4.3.1 and 4.5.1.

3.6.1 Misfires. The refurbished cartridges shall not fail to fire from the weapon when they are initially dropped into it.

3.6.2 Short rounds. No refurbished cartridges shall have a horizontal range less than 80 percent of the average horizontal range (calculated without the short rounds).

3.6.3 Metal parts separation. There shall be no metal parts separation in the gun bore or in flight.

3.6.4 Range and deflection precision. The refurbished cartridge shall have a horizontal range probable error (PEr) less than 10 meters for all zones. The probable error deflection (PEd) shall be less than 10 meters for all zones.

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3.6.5 Duds. The fuze shall function upon ground impact of the refurbished cartridge.

3.6.6 Premature burst. There shall be no burst in the gun bore or in the first 15 meters of flight.

3.6.7 Weapon. The refurbished cartridge shall not be the cause of weapon damage.

3.6.8 Fuze signature. The fuze, when ballistically fired, shall provide both a visible and audible signature at a minimum range of 550 meters from the observer. Only a visible or only an audible signature upon impact is not sufficient to meet this requirement. Visual and audio enhancement equipment shall not be used for this assessment. Visible is defined as flash or smoke or both flash and smoke.

3.6.9 Dud plug function.

3.6.9.1 Functioned round. All dud plugs shall blow out of the projectile body upon fuze function.

3.6.9.2 Inert round. No dud plug shall exit the projectile body during ballistic flight or upon impact.

3.7 First article . This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor are specified in the contract and by 4.3 herein.(See 6.2)

3.8 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and any other foreign matter. Any cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

3.9 Dud plugs.

3.9.1 Configuration. Three dud plugs shall be present and properly inserted into the projectile body for the proving ground test.

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3.9.2 Compliance. The dud plugs shall comply with the requirements of QAP C12961001. The contractor shall certify to that compliance in one of the following ways:

a. Incoming inspections and testing in accordance with the QAP were performed in his facilities.

b. Incoming inspections and testing in accordance with the QAP were performed at another facility approved by the Government.

c. The dud plugs were purchased with the QAP as part of the procurement documentation.

d. The QAP may be applied to the contractor that packs out the M776 Refurbishment Kit. In this case, the dud plugs shall be found acceptable by the Government before releasing the inspection lot quantity to the contractor for use in production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. The requirements of MIL-A-48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item.

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 and the first article ballistic test shall be rejected if:

a. Two or more misfires occur.

b. Three or more duds occur without any misfires or if two or more duds occur with one misfire.

c. One or more short rounds occur.

d. One or more premature bursts occur.

e. One or more metal parts separation occur.

f. Precision requirements are not met. (70 degrees F conditioned samples only).

g. One or more dud plugs exit the projectile body in the inert fuze ballistic test.

h. Two or more dud plugs fail to exit the projectile body in the live fuze ballistic tests (accumulatively, except in the case of a particular round that was an actual dud).

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-K-71151 (AR)

PARAGRAPH	TITLE Packing and Marking of Box, Fiberboard, Inner for Refurbishment Kit, M766	SHEET 1 OF 2		DRAWING NUMBER See below NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Packing and Marking of Box, Fiberboard, Inner for Refurbishment Kit, M766 (Dwg. 12953609)			
	Examination for defects	10	3.1	4.4.2.6
	Proving ground test	96	3.6	Test/4.5.1
	Plug, Increment (Dwg. 12944200)			
	Examination for defects	10	3.1	4.4.2.5
	Increment plug pull test	10	3.5	Test/4.5.2
	Pin, Firing (Dwg. 12944190)			
	Examination for defects	10	3.1	4.4.2.1

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Packing for Box, Fiberboard, Inner for Refurbishment Kit, M766	SHEET 2 OF 2		DRAWING NUMBER See below NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Plug, Breech (Dwg. 12953572) Examination for defects	10	3.1	4.4.2.2
	Breech Plug and Firing Pin Assembly (Dwg. 12953571) Examination for defects	10	3.1	4.4.2.3

NOTES:

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of kits shall contain:

a. Propelling charges made from not more than one inspection lot of propellant, from one manufacturer.

b. Obturating rings of one inspection lot, from one manufacturer.

c. Fuzes of one inspection lot, from one manufacturer.

d. Breech plug and firing pin assemblies from one lot inspection, from one manufacturer.

e. Dud plugs from one inspection lot that comply with 3.9.2.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the Classification of Characteristics paragraphs which follow. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes Sampling Inspection shall be conducted in accordance with TABLE II below, using the inspection levels cited in the Classification of Characteristics paragraphs:

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TABLE II. Attributes sampling inspection

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	13	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

(Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.)

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Pin, Firing	1 OF 1	12953572
			NEXT HIGHER ASSEMBLY 12944192

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total length	III	3.1	Gage
102	Large outside diameter	III	3.1	Gage
103	Small outside diameter	III	3.1	Gage
104	True position of small outside diameter to large outside diameter	III	3.1	Gage
105	Length of large outside diameter section	III	3.1	Gage
106	Length of firing pin from large outside diameter face	III	3.1	Gage
107	True position of firing point to large outside diameter	III	3.1	Gage
108	Firing point radius	III	3.1	Gage
Minor				
201	Improper surface finish	V	3.1	Visual
202	Protective finish missing or incomplete	V	3.1	Visual
203	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Plug, Breech	1 OF 1	12953572
			NEXT HIGHER ASSEMBLY 12944192

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Large inside diameter	III	3.1	Gage
102	Small inside diameter	III	3.1	Gage
103	Depth of large inside diameter	III	3.1	Gage
104	True position of large inside diameter to small inside diameter	III	3.1	Gage
105	True position of small inside diameter to outside thread	III	3.1	Gage
106	Outside thread major diameter, min.	III	3.1	Gage
107	Total length of plug	III	3.1	Gage
108	Pitch diameter of thread	III	3.1	Gage
Minor				
201	Outside diameter of plug	V	3.1	Gage
202	Length of outside diameter from front face to thread	V	3.1	Gage
203	Diameter of spanner wrench holes	V	3.1	Gage
204	Depth of spanner wrench holes	V	3.1	Gage
205	Location of spanner wrench holes	V	3.1	Gage
206	Protective finish missing or incomplete	V	3.1	Visual
207	Improper surface finish	V	3.1	Visual
208	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12953571
4.4.2.3	Breech Plug and Firing Pin Assembly		NEXT HIGHER ASSEMBLY 12953536

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Firing pin not free to move	100%	3.1	Manual
102	Stake protrusion inadequate	III	3.1	Gage
103	Firing pin does not exceed min. projection dimensions	III	3.1	Gage
Minor				
201	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Plug, Increment	1 OF 1	12944200
			NEXT HIGHER ASSEMBLY 12953536

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of plug insert	(a)	3.1	Gage
102	Length of plug insert	(a)	3.1	Gage
103	Thickness of plug tab	(a)	3.1	Gage
104	Width of pull tab	(a)	3.1	Gage
105	Thickness of pull tab to insert connector	(a)	3.1	Gage
106	Width of pull tab to insert connector	(a)	3.1	Gage
107	Increment plug pull test	V	3.3.5	Test/4.5.2
Minor				
201	Outside diameter of insert dome	(a)	3.1	Gage
202	Length of pull tab	(a)	3.1	Gage
203	Poor workmanship	V	3.8	Visual

NOTES: (a) 20 samples from each mold cavity shall be examined for this feature or dimension. The sample of 20 shall include the first two parts from the beginning of each production run and the last two parts from the end of the run. The remaining sample of (at least 16) shall be randomly selected during the production run. Any defects found in any of the 20 samples from each die cavity shall be cause for rejection of all increment plugs produced from that particular die cavity.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Packing and Marking of Box, Fiberboard, Inner for Refurbishment, Kit M766 (while packing)	1 OF 1	12953609
			NEXT HIGHER ASSEMBLY 12953610

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special A	Red marking visible on fuze striker	100%	3.1	Visual
Major 101	Kit components improperly packed	III	3.1	Visual
102	Kit components missing	III	3.1	Visual
Minor 201	Poor workmanship	V	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.6	TITLE Packing and Marking of Box, Fiberboard, Inner for Refurbishment, Kit M766 (after packing)	SHEET 1 OF 1		DRAWING NUMBER 12953609
				NEXT HIGHER ASSEMBLY 12953610
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged	III	3.1	Visual
102	Contents loose	III	3.1	Visual/Manual/ Tactile
103	Sealing tape improperly applied or badly wrinkled	III	3.1	Visual
Minor				
201	Marking missing, misleading or unidentifiable	V	3.1	Visual
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Fiberboard, Outer for Refurbishment, Kit M766 (while packing)	1 OF 1	12953610
			NEXT HIGHER ASSEMBLY 12953611

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wrong number of kits in box	III	3.1	Visual
102	Tape improperly applied to box	III	3.1	Visual
103	Barrier bag in properly placed or sealed	III	3.1	Visual
Minor				
201	Marking missing, misleading or unidentifiable	V	3.1	Visual
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Packing and Marking for Box, Wood, Wirebound for Refurbishment, Kit M766 (while packing)	1 OF 1	12953611
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Damaged barrier bags	III	3.1	Visual
Minor 201	Spacers or fillers inadequate to prohibit movement of contents	V	3.1	Manual/Tactile
202	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.9	TITLE Packing for Box, Wood, Wirebound for Refurbishment, Kit M766 (after packing)	SHEET 1 OF 1		DRAWING NUMBER 12953611
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Boards split or loose, exposing box's contents	V	3.1	Visual
202	Box's contents loose	V	3.1	Visual/Tactile/Manual
203	Marking missing, misleading or unidentifiable	V	3.1	Visual
204	Box wire locking tabs insecure	V	3.1	Visual/Manual
205	Seal missing, unsealed or improperly positioned	V	3.1	Visual/Manual
206	Poor workmanship	V	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.10	TITLE Packing and Marking of Box, Fiberboard, Inner for Refurbishment, Kit M766 (after refurbishing a spent cartridge)	SHEET 1 OF 1		DRAWING NUMBER 12953609 / NEXT HIGHER ASSEMBLY 12963610
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
A	Red marking visible on fuze striker	(a)	3.1	Visual
B	Fuze striker in armed position	(a)	3.1	Visual
C	Fuze packing clip missing or improperly assembled	(a)	3.1	Visual
D	Obturating ring missing	(a)	3.1	Visual
E	Breech plug assembly firing pin not below flush after assembly to spent cartridge	(a)	3.1	Visual
F	Powder from energetic cartridges found on outside of round due to poor seals on shotgun shells	(a)	3.1	Visual
G	Any dud plug(s) missing	(a)	3.9.1	Visual
Major				
101	Obturating ring not flush or below bourrelet diameter	(a)	3.2	Gage
102	Firing pin does not move freely within breech plug	(a)	3.1	Manual/Tactile
103	Increment plug pull test	III	3.5	Test/4.5.2
104	Proving ground test	See4.4.3.1 (b)	3.6	Test/4.5.1
Minor				
201	Poor workmanship	V	3.8	Visual

NOTES:
(a) All test quantities.
(b) The proving ground test shall not be performed prior to the successful completion of all examinations for defects and testing.

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4.4.3 Testing.

4.4.3.1 Proving ground test.

4.4.3.1.1 Observation criteria. The refurbished cartridges shall be observed for the following characteristics:

a. Misfires, major defect (see 3.6.1)

b. Duds, major defect (see 3.6.5)

c. Short rounds, special defect (see 3.6.2)

d. Premature burst, critical defect (see 3.6.6)

e. Metal parts separation, special defect (see 3.6.3)

f. Precision, major defect (see 3.6.4)

g. Weapon damage, major defect (see 3.6.7)

h. Proper impact signature, major defect (see 3.6.8)

i. Improper dud plug function, major defect (see 3.6.9)

4.4.3.1.2 Sampling and acceptance criteria. Testing shall be as specified in 4.5.1. The lot shall be rejected if in sections a, b and c of 4.5.1.1 combined:

a. Two or more misfires occur.

b. Three or more duds occur without any misfires or if two or more duds occur with one misfire.

c. One or more short rounds occur.

d. One or more premature bursts occur.

e. One or more metal parts separation occur.

f. Precision requirements are not met, at ambient temperature (70 degrees F).

g. If the weapon is damaged by abnormal wear of firing pin, or caused by excessive pressure.

h. Two or more improper impact signatures.

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i. One or more dud plugs exit the projectile body in the inert fuze ballistic test.

j. Two or more dud plugs exit the projectile body in the live fuze ballistic tests (accumulatively, except in the case of a particular round was an actual dud).

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Proving ground ballistic test.

4.5.1.1 Test group and firing criteria. The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure. The projectile bodies and fin bodies (both of which can be previously fired up to a total of 5 times) shall be used for this test. The refurbished cartridges shall be fired from a standard 60MM M224 mortar, baseplate, at an elevation of forty-five (45) degrees of 800 mils. The refurbished cartridges shall be grouped as follows:

a. One group of 20 each conditioned for at least 24 hours at +110 degrees F $\pm$ 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3.

b. Two groups of 16 each conditioned for at least 24 hours at 70 degrees F $\pm$ 2.5 degrees F (see 4.4.3.1.2), one group shall be fired at charge 0 and the other at charge 3.

c. One group of 20 each conditioned for at least 24 hours at 0 degrees F $\pm$ 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 0.

d. One group of 24 each with inert fuzes conditioned for at least 24 hours at 0 degrees F +/- 2.5 degrees F (see 4.4.3.2.1), shall be fired at charge 3.

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All of the above refurbished cartridges shall be fired within two minutes after removal from the conditioning chamber. A smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just out of the gun muzzle. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all other rounds in that particular charge zone, in that lot, shall be attached to the firing record. Film from acceptable lots may be discarded. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. Film from acceptable lots may be discarded. Should metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The individual muzzle velocities, deflections and ranges shall be recorded on the firing record. Range probable error shall be calculated and recorded on the firing record for groups a, b and c above.

4.5.1.2 Determination of safe fuze condition.

4.5.1.2.1 External fuze examination. Before removal of the ogive, the red band on the striker which indicates an armed fuze must not be visible. Similarly, loose powder from the spotting cartridge must not be visible. If either or both of these conditions are found, the fuze shall be considered unsafe to handle.

4.5.1.2.2 Damaged fuze. A broken fuze shall be considered safe to dispose of providing that the spotting cartridge is still intact and the striker half of the fuze can no longer come in contact with the spotting cartridge.

4.5.2 Increment plug, pull test. The increment plugs shall be removed from the projectile body by a direct pull normal to the projectile surface and gaged by pull measuring devices simultaneously. Observation shall be made for the force required to remove the plug. If any of the test sample units fail to meet the pull requirement limits, that sample unit shall be classified defective. As an alternative method, an approved gaged can be substituted for the projectile body.

4.5.3 Test validity. If for any reason the ARDEC considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

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5. PACKAGING

5.1 Unit packaging. Unit packaging for refurbishment kit shall be in accordance with requirements of Drawing 12953609.

5.2 Intermediate packaging. Intermediate packaging for refurbishment kit shall be in accordance with requirements of Drawing 12953610.

5.3 Packing. Packing for refurbishment kit shall be in accordance with requirements of Drawing 12953611.

5.4 Marking. Marking shall be as specified on Drawings 12953609, 12953610 and 12953611.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The item covered by this specification is intended for use in Cartridge, 60mm, Practice Short Range, M766.

6.2 Acquisition requirements.

a. See MIL-A-48078.

b. Issue of DODSS to be cited in the solicitation and if required, the specific issue of individual documents referenced (See 2.1).

c. Provisions for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.5 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to the following: Commander, ARDEC, ATTN: SMCAR-QAT-B, SMCAR-FSS-DM, and SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.8 Subject term (key word) listing.

Ballistic tests
Deflection precision
Pull test

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AE15)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-K-71151 (AR)	2. DOCUMENT DATE (YYMMDD) 931101

3. DOCUMENT TITLE: KIT, REFURBISHMENT FOR CARTRIDGE, 60MM, PRACTICE, SHORT RANGE M766 LAP

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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